# TARUN KARNATI

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## SUMMARY

Software Developer with hands-on experience building and deploying real-world applications across backend, cloud, and AI-driven solutions. Strong problem-solving and analytical mindset with experience working in collaborative, fast-paced environments. Adaptable, quick to learn new technologies, and open to exploring different software development opportunities.

## TECHNICAL SKILLS

**AI/ML:** MCP, A2A, LLM Integration, RAG, LangGraph, LangChain, Computer Vision, Vector Embeddings, Prompt Engineering

**AI APIs & Libraries:** Claude API, OpenAI API, Gemini API, Hugging Face, scikit-learn, TensorFlow, PyTorch, Pandas, NumPy

**Languages:** C, Java, Python, JavaScript, HTML/CSS, SQL, Bash

**Cloud & DevOps:** Microsoft Azure, Google Cloud (Vertex AI, Cloud Run), Firebase, Docker, CI/CD

**Frameworks**: React, Node.js, Express.js, Flask, FastAPI, RESTful APIs

**Databases:** MongoDB, MySQL, Firestore, ChromaDB

**Tools:** Git, GitHub, Postman, Jira, Render, Netlify

## PROFESSIONAL EXPERIENCE

**Generative AI Engineer, Team Lead** | EvenHub Edu, AI-Powered Education Platform
*April 2026 to Present* | *Toronto, ON*

- Built a context-aware AI assistant on Gemini 2.5 Flash with a dual-source RAG pipeline over school documents and parent Gmail (OAuth 2.0), streamed over SSE with function-calling tools.
- Added a Vertex AI reranking stage over 30 candidate chunks with score-based trimming, improving answer relevance with no re-indexing.
- Architected a multi-tenant platform with 4-tier RBAC, isolating tenants in the AI layer by scoping the chat context cache per role and school.
- Containerized and deployed a Node.js and Express backend to Google Cloud Run via GitHub Actions, cutting costs by 70%.

## PROJECTS

**BakeOps Command Center, Multi-Agent AI for Industrial Operations**
*2nd Prize, TMLS Agentic Hackathon x FGF Brands* | *Python, LangGraph, Claude API, MCP, A2A, AG-UI, FastAPI*

- Led a team building a hierarchical multi-agent system on a LangGraph supervisor graph, with a Claude Opus orchestrator routing tasks to Sonnet specialist agents across maintenance, supply chain, quality, and scheduling.
- Wrapped an external product-data API as an MCP server and invoked it through an MCP client, giving agents tool access over the protocol.
- Streamed agent reasoning, tool calls, and telemetry to a React dashboard over SSE using the AG-UI event protocol for real-time traceability.

**Compass, Enterprise RAG Agent for Microservice Documentation**
*Python, LangGraph, Qdrant, Cohere, Claude, Langfuse, MCP, Slack Bolt*

- Built a Slack agent answering engineering questions grounded strictly in OpenAPI specs and internal docs, with inline citations across a 12-service corpus.
- Designed hybrid retrieval prefetching 25 dense and 25 sparse candidates, fused with Reciprocal Rank Fusion and reranked to 8 with Cohere.
- Built a LangGraph agent graph that classifies intent, extracts metadata filters, rewrites follow-ups, grades retrieved context, and retries before generating.
- Built a 50-question evaluation harness measuring retrieval quality at 97.6% hit rate and 0.938 MRR, with every agent step and tool call traced in Langfuse.

**ORCA AI – AI-POWERED DEVOPS & AUTOMATION PLATFORM**
*Python, FastAPI, MCP , Multi-LLM, Playwright*

- Architected an autonomous DevOps platform using MCP to orchestrate AI agents across 4 domains: browser testing, GitHub automation, shell execution, and cloud deployment in a single unified pipeline.

- Engineered multi-model AI orchestration across 3 providers (OpenAI, Anthropic, Google) with intelligent fallback routing, ensuring zero-downtime inference by auto-switching on failure or latency spikes.
- Implemented a self-healing CI pipeline where AI agents detect, diagnose, and auto-fix code errors, enabling fully autonomous prompt-to-deployment delivery with zero manual intervention.

## CERTIFICATIONS

- Introduction to Model Context Protocol, Anthropic
- Microsoft AZ-900: Azure Fundamentals, Microsoft
- Docker Foundations Professional Certificate, Docker and LinkedIn Learning
- Generative AI: Working with Large Language Models, NASBA and LinkedIn Learning
- Career Essentials in Software Development, Microsoft and LinkedIn

## EDUCATION

**Postgraduate Diploma, Information Technology Solutions** | Humber College, Toronto, ON | Jan 2025 to Aug 2026

**Bachelor of Technology, Data Science** | SRK Institute of Technology, India | 2020 to 2024

## PAPER PUBLICATIONS

Public Relationships Using NLP and Machine Learning. IJCRT, Volume 12, Issue 5, May 2024 (ISSN: 2320-2882).

## ACHIEVEMENTS AND AWARDS

- TMLS Agentic Hackathon 2026, 2nd Prize, Toronto Machine Learning Society x FGF Brands.
- Brainstem Alliance Hackathon 2025, 2nd Prize, Community Code Days.
- Dean's Honour List, Humber College, semesters 1, 2, and 3.